

Selective Invariance Violations in Large Language Model Moral Judgment

A Geometric Framework for Behavioral Manipulation Detection

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The problem: LLMs as decision services

- LLMs increasingly gate **safety-critical decisions** — content moderation, clinical triage, legal analysis.
- A secure evaluator should judge the **facts**, not their **presentation**.
- **Behavioral manipulation vulnerability**: an attacker shifts the decision by changing morally *irrelevant* surface features — framing, tone, sensory detail — without changing the underlying situation.

The measurement gap

Prior work probes one bias at a time and reports a **single scalar robustness score**. Across n independent failure directions, a scalar destroys $n - 1$ of them. **No post-hoc procedure recovers the lost structure.**

Our approach: a geometric vulnerability profile

- Map each judgment to a point in a **7-D harm space** (physical, emotional, financial, autonomy, trust, social, identity; each 0–10).
- Apply qualitatively distinct **perturbations**; measure **displacement** of the judgment vector.
- Output a **per-model vulnerability profile**, not one number.
- Runs under a realistic **\$50/day** compute budget.



The 7-D space is a measurement-reliable projection of the **DEME v3** 9-D moral vector.

Grounding: 7-D harm space → DEME v3 9-D vector

Harm dim.	DEME v3 axis	Mapping
physical	physical harm	direct
autonomy	autonomy / consent	direct
trust	legitimacy / trust	direct
social	societal / environmental	direct
financial	fairness / equity	partial
emotional	virtue / care	partial
identity	privacy / standing	partial
—	rights, epistemic quality	not scored

The evaluation pipeline is the **measurement front-end**; the ErisML compiler's **DEME bridge** is the downstream decision kernel (DEMEVerdict).

Finding 1 — Vulnerabilities are *selective*

Displace judgments (real attack surface):

- **Linguistic framing**
- **Emotional anchoring** ($d = 0.6\text{--}1.1$)
- **Irrelevant sensory detail**

Do not displace:

- Gender swap
- Evaluation order

We lead with **effect sizes** (Cohen's d , displacement in harm-points) over σ ; empirical control arms rule out stochastic drift.

The common thread

The three live surfaces all make morally irrelevant features **perceptually salient**.

⇒ The attack mechanism is **salience manipulation** — and the framework discriminates it from noise.

Finding 2 — Robustness profiles are dissociable

- **No model dominates** all attack surfaces.
- Claude: **zero sycophancy** — but **worst** emotional-anchoring recovery (20%) and worst divided attention.
- Flash 2.0: **best** anchoring recovery (73%) — but **worst** working memory.

Consequence for Secure AI

Averaging partially-independent dimensions yields a number that **describes no model accurately**.

Single-test certification provides false assurance.

Is the 7-D space reliably measurable, or asserted?

Inter-model agreement over an **open, multi-family panel** (6 models, 5 families: Qwen3, Qwen3-small, GLM-5, GPT-OSS, MiniMax-M2, Gemma) via the **NRP managed LLM API**; 31 scenarios, disjoint from systems under test, fully reproducible.

	ICC(2, k)	Krippendorff α
Financial (highest)	0.983	0.902
Trust	0.958	0.783
Physical / Autonomy (lowest)	0.893	≈ 0.57
Overall	0.969	0.836

Test-retest median $r = 0.96$. Dimensions index a shared, model-independent structure.

Content-moderation filter: flag if total harm $> T$.

- Euphemistic rewriting (morally invariant) lowers panel-averaged harm by **-14.0 points** on the 0–70 scale (4/6 gold > 10 pts); dramatic only +7.3.
- At a median-calibrated threshold, **3 of 6** gold items silently reclassify FLAG→PASS.
- **Asymmetry**: far easier to *evade* (hide harm) than to *trip* a false positive — the dangerous direction.

NEW — The exploit survives a principled kernel

Route each scenario through the real **ErisML DEME v3** decision kernel:

text → **LLM extracts ethical facts** → GenevaEMV3 → typed verdict
(FORBID → AVOID → NEUTRAL → PREFER → STRONGLY_PREFER)

Across 161 scenario–model cases

- Euphemistic rewriting flips FORBID/AVOID→permissive in **13.7%** of cases; mean shift **+0.65** ordinal steps.
- FORBID verdicts fall **44 → 27** (−39%); STRONGLY_PREFER rises 46 → 78.

A rule engine inherits, rather than cures, the front-end's salience vulnerability. Harden *fact extraction*, not just the rule.

Since submission — a measured defense

Equivalence-class averaging: generate the input's paraphrase class (m paraphrases), **average per-dimension perception over the class**, then decide.

- Pre-registered bar: decision-layer drift $\theta_d \leq 0.5$. At scale (60 held-out items, $m = 6$): raw drift **0.407** $\rightarrow$ **0.219** — the mechanism **halves** salience drift and meets the bar.
- LLM paraphraser **refuse 24%** of harmful inputs. A non-refusing red-team paraphraser (NLLB back-translation, 6 pivots): $\theta_d = 0.301$ on harmful content — the hole is the **generator's**, not the mechanism's.
- Trust boundary: refusal $\rightarrow$ singleton class $\rightarrow$ **escalate by default**; audit proof records class members.

Natural (not adversarial) paraphrases at scale. Post-camera-ready results, July 2026 — extended version in preparation.

`pip install moral-spectrum-analyzer`

Since submission — the harm space is extensible *and falsifiable*

Discovery loop: residual analysis flags candidate missing dimensions → pre-registered admission gate.

- Scorecard: 3 flagged → **1 validated** (identity_attack), **1 retracted** (threat), **1 declined** (sexual content → policy channel, not a moral axis). **The gate has teeth.**
- identity_attack: cross-dataset held-out AUROC **0.80** CI [0.78, 0.83] (+0.25 over null, 2 corpora, $n = 6400$) — wired live as a **10th channel**; dominant feature of a learned moderation contraction (OOF AUROC **0.863**, leakage-controlled).
- Cross-lingual at scale: invariance index **0.72–0.80** across es/ar/zh/hi/sw, **harmful $\approx$ benign**.

Validated on the learned-encoder perception layer (xbse); native DEME re-run of the five tracks is the extended version.
Post-camera-ready results, July 2026.

Implications for Secure AI deployment

- **Where to look:** attacks that repackage the same facts with different **salience** (euphemistic minimization).
- **Prompt-level defenses are bounded:** explicit warnings recover only $\sim 38\%$ — a ceiling co-occurring with universal overconfidence (ECE 0.19–0.42).
- **Ask the right question:** “*which* vulnerabilities matter for our use case?” — not “what’s the robustness score?”
- **Hardening target:** the fact-extraction front-end, since downstream kernels inherit its failures.

Reproducibility & takeaways

- Five tracks, multi-model, ~8,000 calls — under **\$50/day** (Kaggle); validation panel on the open **NRP managed LLM API**.
- Code + data: `pip install agi-hpc`, tag `bds2026-v1`; decision kernel: **ErisML / DEME v3**.

Four takeaways

- ① Vulnerabilities are **selective** — salience manipulation is the surface.
- ② Robustness profiles are **dissociable** — no single score is safe.
- ③ The exploit is **end-to-end** — it flips scalar scores *and* typed verdicts.
- ④ It is **defensible** — averaging over the paraphrase class halves drift (since submission).



Thank you

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Questions?